

Chapter 14: `upper_bound()`

If you understand `lower_bound()`, then `upper_bound()` is easy.

Just memorize:

```
lower_bound(x) -> first element >= x
upper_bound(x) -> first element > x
```

That one line solves most confusion.

What Does `upper_bound()` Do?

Definition:

Returns an iterator to the **first element strictly greater than x**.

```
upper_bound(x)
=
first element > x
```

Example 1

```
1 3 5 7 9
```

Find:

```
upper_bound(5)
```

Answer:

```
7
```

Because:

```
7 > 5
```

and it is the first such element.

Example 2

```
1 3 5 7 9
```

Find:

```
upper_bound(4)
```

Answer:

```
5
```

Example 3

```
1 3 5 7 9
```

Find:

```
upper_bound(9)
```

No element > 9

Returns:

```
end()
```

Header

```
#include <algorithm>
```

Syntax

```
upper_bound(begin,end,x);
```

Returns:

```
iterator
```

Basic Example

```
vector<int> v={1,3,5,7,9};

auto it=upper_bound(
    v.begin(),
    v.end(),
    5
);

cout<<*it;
```

Output:

```
7
```

Index

```
cout<<it-v.begin();
```

For:

```
upper_bound(5)
```

Output:

```
3
```

Because:

Index:

```
0 1 2 3 4
1 3 5 7 9
      ^
```

Duplicate Example (Most Important)

Input:

```
1 2 2 2 2 5 7
```

Find:

```
upper_bound(2)
```

Result:

```
5
```

Index:

```
5
```

Visualization:

```
1 2 2 2 2 5 7
      ^
```

Compare Lower vs Upper

Input:

```
1 2 2 2 2 5 7
```

lower_bound(2)

```
1 2 2 2 2 5 7
      ^
```

Index:

```
1
```

upper_bound(2)

```
1 2 2 2 2 5 7
      ^
```

Index:

```
5
```

Count Occurrences (Very Important)

Input:

1 2 2 2 2 5 7

Count number of:

2

Formula:

$\text{upper_bound} - \text{lower_bound}$

Code:

```
int cnt =  
upper_bound(v.begin(), v.end(), 2)  
-  
lower_bound(v.begin(), v.end(), 2);
```

Output:

4

This is one of the most common interview tricks.

First Occurrence

`lower_bound(x)`

gives first occurrence.

Input:

1 2 2 2 2 5

```
lower_bound(2)
```

returns:

```
1
```

(index)

Last Occurrence

Formula:

```
upper_bound(x) - 1
```

Example:

```
1 2 2 2 2 5
```

```
upper_bound(2)
```

returns index:

```
5
```

Last occurrence:

```
5 - 1 = 4
```

Correct.

Using With Set

```
set<int> s={10,20,30,40};  
  
auto it=s.upper_bound(20);  
  
cout<<*it;
```

Output:

30

Using With Map

```
map<int,string> mp;  
  
mp[10]="A";  
mp[20]="B";  
mp[30]="C";
```

```
auto it=mp.upper_bound(20);
```

Points to:

30 -> C

Access:

```
it->first  
it->second
```

lower_bound vs upper_bound

Function	Meaning
lower_bound(x)	First $\geq x$
upper_bound(x)	First $> x$

Example:

1 2 2 2 5

For:

x=2

Lower:

1 2 2 2 5
^

Upper:

1 2 2 2 5
^

Common Interview Uses

Count Occurrences

```
count =
upper_bound - lower_bound
```

First Occurrence

```
lower_bound
```

Last Occurrence

```
upper_bound - 1
```

Range Query

Find all elements equal to x.

Range:

```
[  
  lower_bound(x),  
  upper_bound(x)  
)
```

Complexity

```
O(log n)
```

Practice Questions

Easy

Q1

Find upper_bound of:

```
1 3 5 7 9
```

for:

5

Output:

7

Q2

Print index of upper_bound(3).

Q3

Count occurrences of 2.

Input:

1 2 2 2 5

Output:

3

Q4

Find first occurrence.

Q5

Find last occurrence.

Medium

Q6

Find frequency of x in sorted array.

Q7

Count numbers $\leq x$.

Hint:

upper_bound

Example:

1 3 5 7 9

x:

5

Answer:

3

Q8

Count numbers $< x$.

Hint:

lower_bound

Q9

Find nearest greater element.

Q10

Find range of equal elements.

Interview Level

Q11

Find First and Last Position

LeetCode #34

Q12

Count Occurrences in Sorted Array

Q13

Search Insert Position

LeetCode #35

Q14

K Closest Elements

LeetCode #658

Common Mistakes

Mistake 1

Using unsorted array.

```
5 1 7 3
```

✗ Wrong.

Mistake 2

Dereferencing end().

```
*it
```

when:

```
it==v.end()
```

✗ Runtime Error.

Mistake 3

Confusing definitions.

Remember:

```
lower_bound = >=
```

```
upper_bound = >
```

Revision Sheet

Header

```
#include <algorithm>
```

Syntax

```
upper_bound(  
begin,  
end,  
x  
);
```

Meaning

```
First Element > x
```

Return

```
iterator
```

Complexity

```
O(log n)
```

Formula

Count Occurrences:

```
upper_bound(x)  
-  
lower_bound(x)
```

First Occurrence

```
lower_bound(x)
```

Last Occurrence

```
upper_bound(x) - 1
```

Golden Interview Formula

For a sorted array:

```
frequency(x)  
=  
upper_bound(x)  
-  
lower_bound(x)
```

This single formula appears in many coding interviews.

Next Chapter

`equal_range()`

It combines:

```
lower_bound()  
+  
upper_bound()
```

in a single call and is useful for range queries and duplicate counting.